RadarMAE: Injecting Physical Inductive Biases into Masked Autoencoders for Advanced Radar Object Detection

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Abstract

Masked Autoencoders (MAEs) have emerged as a powerful self-supervised paradigm in vision, yet their direct application to radar perception remains challenged by a fundamental domain gap. Specifically, while conventional MAEs are designed for RGB images using isotropic grid patching and standard mean squared error (MSE) loss, radar range-azimuth (RA) heatmaps exhibit anisotropic spatial structures and axis-specific physical artifacts, including multipath reflections and sidelobe responses. In this paper, we propose **RadarMAE**, a radar-aware MAE designed to incorporate the physical characteristics of radar signals. We introduce an axial tokenization strategy that accommodates the resolution disparities between the range and azimuth axes while capturing spatially extended radar responses. Furthermore, considering the distinct characteristics of the two axes, we introduce auxiliary distribution-based losses that encourage consistency in range- and azimuth-profile intensity statistics during masked reconstruction. Using only single-frame RA maps, RadarMAE achieves 64.17% AP_{BEV} at IoU 0.3 on K-Radar-20, outperforming vanilla MAE by 7.29 AP points and its architecture-matched scratch baseline by 3.40 AP points. These results support the effectiveness of incorporating radar-specific physical priors into self-supervised representation learning, yielding representations that benefit downstream radar object detection.[2]

1 Introduction

Frequency-modulated continuous-wave (FMCW) radar plays a pivotal role in autonomous driving (AD) systems, primarily due to its reliable perception capabilities even in adverse weather conditions that severely degrade cameras and LiDAR [1–3]. To extract actionable targets from the radar signals, traditional radar detection algorithms have leveraged signal processing techniques such as the constant false alarm rate (CFAR) detector [4, 5]. However, this approach causes severe information sparsification due to its reliance on a heuristic thresholding pipeline, inevitably discarding weak signal reflections and crucial geometric semantics [6–9]. To overcome this limitation, a paradigm shift toward deep learning-based radar object detection has emerged [1, 2, 6, 10, 11], where supervised learning schemes are generally utilized to capture rich spatial contexts and subtle target signatures directly from dense radar heatmaps. Leveraging these advantages, data-driven radar perception methods can preserve richer signal information and achieve superior detection performance compared to conventional CFAR-based pipelines [2, 6, 12–14].

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[2] Project page: https://gracenpraise.github.io/RadarMAE/

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Despite these advancements, fully unlocking the potential of the radar domain remains bottlenecked by the intrinsic limitations of supervised learning, specifically its strict reliance on massive, high-quality annotations corresponding to input radar heatmaps. Compared to camera or LiDAR data, radar heatmaps are difficult to interpret visually. This non-interpretability stems from the superposition of target returns with multipath reflections and background clutter, further aggravated by sidelobe artifacts that manifest as ghost replicas of the true response. Such intertwined physical phenomena obscure clear geometric boundaries, thereby making manual annotation exceedingly expensive and error-prone [3, 15]. To address this bottleneck, self-supervised learning (SSL) provides a promising approach [16–19], reducing reliance on manual annotations by learning robust representations from large-scale unlabeled radar data [3, 20–27].

Among the available SSL frameworks [16–19], masked autoencoders (MAEs) [17] stand out as an exceptionally suitable architecture for object detection in radar heatmaps: its dense patch-level reconstruction forces the model to capture fine-grained local spatial structures, which is crucial for understanding the intricate geometric relationships of object-specific sidelobe and multipath components. Motivated by these representational advantages, recent studies have attempted to apply MAEs to unlabeled radar signals [20, 21, 26–28]. However, directly applying vision-centric MAE designs to radar heatmaps may fail to capture the distinctive characteristics required for effective radar representation learning in autonomous driving. This limitation stems from the fundamental differences between camera images and radar heatmaps, as illustrated in Fig. 1(a) and (c). While optical images capture visually interpretable object boundaries and surface appearances, radar RA heatmaps represent the same targets through electromagnetic reflection responses processed into range and azimuth measurements, typically involving Fourier-based signal processing. Given this fundamental domain gap, directly adopting MAE designs developed for natural images introduces two key limitations when learning representations from radar heatmaps.

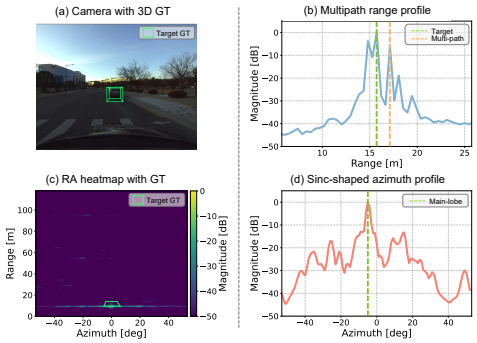


Figure 1: Comparison of camera and radar observations: (a) camera image, (c) corresponding RA heatmap, and radar intensity profiles illustrating (b) range-extended responses and (d) azimuth sidelobes.

The first mismatch lies in the tokenization strategy. Specifically, the standard MAE utilizes square patches, implicitly assuming an isotropic space where the x (*i.e.,* width) and y (*i.e.,* height) axes possess equivalent spatial semantics. In contrast, radar representation, such as range-azimuth (RA) heatmaps, exhibits intrinsically anisotropic properties. Namely, FMCW radar suffers from severe resolution disparity between the range and angle axes. Moreover, target energy disperses along the entire axes in the form of sinc-shaped lobes in polar coordinates (Fig. 3) [9, 29]. Consequently, the spatial footprint of isotropic square patches in the current MAE scheme is inherently incapable of reflecting this asymmetric nature of radar heatmaps as well as fully capturing the sinc-lobe structures.

The second limitation concerns the reconstruction objective. Standard MAE employs a pixel-wise mean squared error (MSE) loss [17], which does not explicitly account for the axis-specific statistical characteristics of radar signals. When masked regions admit multiple plausible reconstructions, MSE-based predictions tend toward the conditional mean, potentially smoothing informative radar responses [30, 31]. In RA heatmaps, multipath reflections can produce delayed and asymmetric intensity profiles along the range axis [32–34], while finite-aperture effects give rise to structured sidelobe patterns along the azimuth axis. These distinct characteristics motivate incorporating axis-specific distributional objectives that encourage the preservation of range- and azimuth-profile statistics during masked reconstruction.

To address these domain-specific challenges, we propose **RadarMAE**, a radar-aware masked autoencoder designed to incorporate inductive biases from radar measurements. Our framework comprises two key components: (1) **axial tokenization**, which represents RA heatmaps as orthogonal, axis-aligned token sequences to accommodate anisotropic resolution and spatially extended radar responses; and (2) **physical distribution-based losses**, which encourage the preservation of axis-specific statistical characteristics associated with multipath reflections and sidelobes during masked reconstruction.

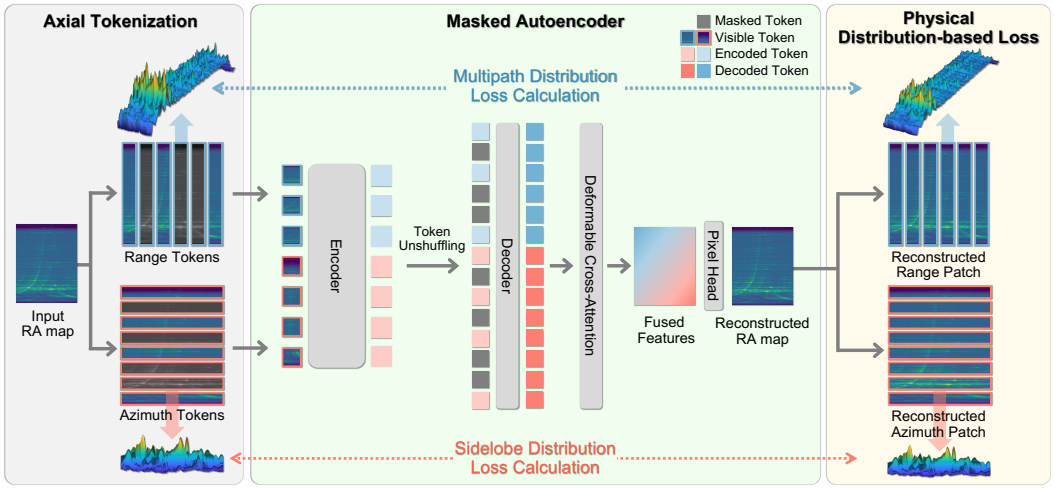


Figure 2: Overview of the RadarMAE pre-training pipeline. RA heatmaps are converted into axial tokens and processed with intensity-weighted masking. Visible tokens are encoded and fused through deformable cross-attention, while reconstruction is guided by pixel-wise and axis-specific physical distribution-based losses.

The main contributions of this work are summarized as follows:

1. We propose axial tokenization, which represents range–azimuth (RA) heatmaps as orthogonal, axis-aligned token sequences. This design explicitly accounts for the anisotropic measurement structure of radar signals while reducing the number of tokens compared with conventional square patch tokenization.
2. We develop a radar-aware masked pre-training strategy that combines intensity-weighted block masking with axis-specific physical distribution-based losses. These components emphasize informative radar responses and encourage the preservation of range- and azimuth-profile statistics during reconstruction.
3. Using only single-frame RA maps, RadarMAE achieves a bird's-eye-view (BEV) AP of 64.17% at IoU 0.3 on K-Radar-20, outperforming vanilla MAE by 7.29 AP points and its architecture-matched scratch baseline by 3.40 AP points.

2 Related Work

Radar Perception with SSL. Unlike camera or LiDAR data, radar signals inherently lack human interpretability, making it exceedingly difficult to construct high-quality, large-scale labeled datasets [3, 15]. To circumvent this critical limitation, recent research has increasingly gravitated toward SSL to exploit vast amounts of unannotated raw radar data [3, 20–24, 26, 27]. Early SSL frameworks in this domain primarily adapted contrastive learning (CL) paradigms, such as SimCLR [16] and MoCo [18], to radar heatmaps [3, 24, 25, 35]. However, CL is fundamentally designed to extract global, instance-level representations; while advantageous for classification, this high-level abstraction obscures fine-grained spatial details and yields suboptimal representations for dense prediction tasks such as object detection, where precise localization is critical [36, 37]. In contrast, MAE [17] formulates dense patch-level reconstruction as a pretext task, compelling the network to recover fine-grained local structures rather than instance-level abstractions [38, 39]. This property aligns naturally with radar heatmap-based detection, where target-specific geometric cues, such as sidelobes and multipath interference, are encoded in delicate spatial patterns that global instance-level contrastive objectives may fail to explicitly preserve [29, 32, 36, 37].

Motivated by the representational advantages of masked modeling, recent work has actively extended MAE to various radar applications. In autonomous driving, frameworks such as Robu-MARC [26] have utilized masked autoencoding to enhance radar-camera fusion and robust 3D perception, while BEV-MAE [27] adopts BEV-guided masking for LiDAR point cloud pre-training and 3D object detection. In the remote sensing domain, models like CROMA [20] and SARMAE [28] have

adopted patch-based masking to extract robust representations from multi-modal and SAR imagery. While these pioneering efforts establish the potential of MAE in the radar domain, their primary focus remains confined to application-level usage rather than fundamentally redesigning the MAE framework itself. Namely, existing radar MAE approaches directly adopt the core mechanics of vision-centric MAE (*i.e.,* tokenization scheme and pixel-wise losses) without radar-specific adaptations. Consequently, the learned representations cannot reflect the anisotropic geometry and dispersion of sinc-shaped sidelobes that fundamentally characterize radar data, leaving substantial room for radar-specific designs.

Injecting Domain-Specific Physical Priors into SSL. Incorporating domain-specific physical priors has become a prevailing strategy for representation learning on signals that inherently deviate from the standard image modality. For instance, in audio and spectrogram analysis, frameworks such as AudioMAE [40] and AST [41] improve representation learning by introducing asymmetric patches tailored to the distinct physical semantics of the time and frequency axes. Similarly, adaptations such as VideoMAE [42] for video and SatMAE [43] for remote sensing incorporate structural priors specific to their respective domains. These successes across diverse modalities collectively demonstrate that injecting the inherent heterogeneity of signal dimensions is crucial for capturing robust features. Despite these advances in related modalities, radar-specific inductive biases remain relatively underexplored in SSL.

3 Preliminary

3.1 Range-Azimuth Representation in FMCW Radar

Physical Basis of the Radar Domain Gap. While camera images capture dense and isotropic spatial reflections in a single snapshot, FMCW radar RA heatmaps are constructed through two orthogonal and physically distinct signal processing pipelines [44, 45]. The range axis is the mapped result of the round-trip time delay of chirp signals reflected from objects. Consequently, the spatial resolution along this axis is fundamentally determined by the sweep bandwidth of the transmitted signal. Conversely, the azimuth axis is derived from the spatial phase differences of the reflected signals received across multiple antenna elements. Accordingly, the angular resolution along this axis is fundamentally determined by the size of the radar's physical or virtual antenna aperture. Originating from fundamentally different physical mechanisms (bandwidth vs. spatial aperture), these two dimensions intrinsically embed highly distinct, dimension-specific physical artifacts into the resulting heatmap, which we explicitly formulate in the following subsection.

3.2 Physical Artifacts in RA Heatmap Distributions

Resolution Disparity. Driven by the aforementioned hardware limitations, the fundamentally different physical mechanisms defining range and azimuth yield severe resolution asymmetry. This structural disparity results in highly anisotropic target profiles across the two dimensions, as can also be observed in Fig. 3.

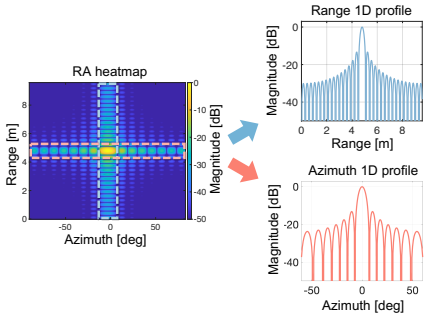


Figure 3: Ideal sinc-shaped point-spread-function (PSF) sidelobes of the FMCW radar RA heatmap.

Multipath Delay. The range dimension exhibits its own distinct artifact: multipath delay, which can distort the range distribution (Fig. 1b) [32–34]. In automotive radar, multipath can arise from indirect reflections involving vehicle surfaces, the ground, and surrounding structures. For example, reflections between a vehicle's undercarriage and the ground surface can produce delayed propagation paths. Because these indirect paths involve longer propagation distances, their responses may appear at farther range bins, contributing to asymmetric and extended range-intensity profiles [32–34]. Although similar multipath effects can also occur in stationary clutter, these target-associated responses may provide additional contextual cues for distinguishing vehicles from surrounding clutter under certain conditions.

Sinc-Shaped Sidelobes. In radar signal processing, applying the fast Fourier transform (FFT) over finite boundaries—restricted by limited bandwidth in range and finite antenna arrays in az-

imuth—mathematically acts as a rectangular window. This inevitably causes spectral leakage in both dimensions, dispersing a target's energy and forming sidelobes governed by $\mathrm{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$ [29, 45, 46]. Similar to multipath effects, these sinc-shaped artifacts form multiple ghost reflections around the target (Fig. 3). Nevertheless, because these sidelobes manifest as deterministic peaks rather than mere random noise, they can aid in understanding the target's underlying properties. Therefore, these inherent sinc characteristics must be explicitly considered when forming effective radar representations.

4 Proposed Methods

To address the limitations of applying standard MAE directly to the radar domain, RadarMAE introduces a radar-aware architecture designed to incorporate the physical characteristics of RA heatmaps. As depicted in Fig. 2, the model is first pre-trained to reconstruct masked radar signals using axial tokenization and radar-aware masking. After pre-training, the decoder is discarded, and a detection head is appended to the pre-trained encoder to fine-tune the network for the downstream task of object detection. The details of each component are as follows.

4.1 Radar-Aware Axial Representation

Axial Tokenization. Standard MAE architectures adopt isotropic square patches, which do not explicitly account for the resolution disparities between the range and azimuth axes or the sinc-shaped reflections extending along these axes. To address this geometric mismatch, we introduce an axial tokenization strategy, which represents the radar signal as orthogonal axial stripes. As illustrated in Fig. 2, by spanning the entire length of each respective axis, these stripes accommodate the resolution disparities while capturing spatially extended radar responses within individual tokens. Prior to linear projection, we apply a lightweight per-stripe early convolution to stabilize local feature extraction without mixing neighboring stripes within each stream. The resulting feature maps ($\tilde{\mathbf{x}}_i^r$ and $\tilde{\mathbf{x}}_j^a$) are then independently projected into range-wise and azimuth-wise tokens:

$$\mathbf{r}_i = \mathbf{W}_r \,\mathrm{vec}(\tilde{\mathbf{x}}_i^r) + \mathbf{b}_r \in \mathbb{R}^d, \tag{1}$$

$$\mathbf{a}_j = \mathbf{W}_a \,\mathrm{vec}(\tilde{\mathbf{x}}_j^a) + \mathbf{b}_a \in \mathbb{R}^d. \tag{2}$$

Here, $\mathrm{vec}(\cdot)$ denotes vectorization, d is the hidden embedding dimension, and $\mathbf{W}_r$, $\mathbf{W}_a$, $\mathbf{b}_r$, and $\mathbf{b}_a$ are learnable projection matrices and bias vectors. Let H and W denote the azimuth and range dimensions, respectively, and P_a and P_r their corresponding tokenization intervals. By decoupling the axes, this formulation reduces the token count from $N_{\mathrm{grid}} = \lceil H/P_a \rceil \lceil W/P_r \rceil$ for conventional grid patching to $N_{\mathrm{axial}} = \lceil H/P_a \rceil + \lceil W/P_r \rceil$, reducing the computational cost of self-attention under the considered configurations.

Physical Positional Encoding. To incorporate radar measurement geometry, we replace conventional index-based embeddings with physical positional encoding (PPE) derived from absolute sensor coordinates. Specifically, PPE maps range coordinates in meters and azimuth coordinates in radians into Fourier features, which are subsequently projected into the hidden dimension through a multilayer perceptron (MLP). After adding PPE and learnable axis-type embeddings to distinguish the two axes, the orthogonal sequences are concatenated into a unified input $\mathbf{Z}_0 = [\mathbf{r}; \mathbf{a}]$. An ablation study comparing PPE with conventional positional embeddings is provided in Appendix G.

4.2 Intensity-Weighted Block Masking

While axial tokenization is highly effective for capturing extended sinc signatures, its orthogonal intersecting structure introduces a critical issue: the potential for local information leakage where the range and azimuth stripes overlap each other. To mitigate this issue and reduce reliance on such shortcuts, we explicitly break this spatial continuity by occluding contiguous segments rather than isolated patches.

Building upon this, we further introduce an intensity-weighted block masking strategy specifically designed for sparse radar RA heatmaps. While standard random masking is highly effective for densely distributed camera images, applying it to radar RA heatmaps risks entirely erasing rare target signals, causing the network to collapse into trivial background interpolation. Based on the domain

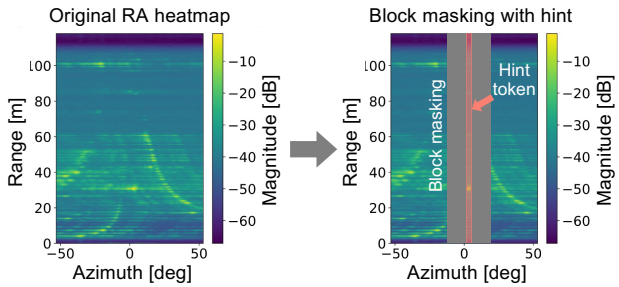


Figure 4: Illustration of intensity-weighted block masking. Contiguous blocks of 5–8 tokens are preferentially selected according to radar intensity. One randomly selected original-measurement token within each selected block remains visible as a hint. Masking is applied independently to the range and azimuth token streams.

prior that high-intensity reflections contain informative target signatures, we prioritize masking these contiguous high-intensity regions to force the network to reconstruct meaningful radar responses rather than background patterns.

Detailed visualizations and ablation analyses of this masking strategy are provided in Appendix F. Finally, to explicitly preserve their distinct physical properties throughout the pipeline, the masked positions are replaced with dedicated learnable mask tokens ($\mathbf{m}_R$, $\mathbf{m}_A$) for the range and azimuth axes, respectively.

4.3 Masked Reconstruction Architecture

Encoder and Decoder. Following the asymmetric MAE paradigm [17], our encoder processes only the visible subset of the unified token sequence $\mathbf{Z}_0$, while a lightweight 8-block decoder reconstructs the signal. Processing the concatenated orthogonal axial stripes through a shared Transformer encoder allows distinct axial inputs to interact via self-attention. This mechanism enables the model to learn cross-axis relationships and geometric dependencies from sparse radar observations. To initiate decoding, we restore the masked positions using the dedicated learnable mask tokens ($\mathbf{m}_R$ and $\mathbf{m}_A$). This asymmetric design encourages the encoder to learn representations from partially observed radar signals.

Deformable Cross-Attention-Based Fusion. To reconstruct the decoupled axial tokens back into a dense 2D patch grid, a geometric mapping is required. We introduce an axial deformable cross-attention mechanism [47]. Let $\mathbf{Q} \in \mathbb{R}^{H_g \times W_g \times d}$ denote a set of learnable 2D spatial queries assigned to a uniform reference grid, where H_g and W_g represent the spatial dimensions. For a query $\mathbf{q}_{ij}$ at reference point p_{ij}, the network predicts continuous 2D sampling offsets Δp_{ijk} and attention weights A_{ijk} for K sampling points. The projected sampling location $p_{ij} + \Delta p_{ijk}$ is subsequently decomposed into orthogonal coordinates x and y. Using these coordinates, the model independently extracts continuous features from the range token bank $\mathbf{V}_r$ and azimuth token bank $\mathbf{V}_a$, which are derived from the outputs of the preceding decoder blocks, via bilinear grid sampling. These sampled features are weighted by A_{ijk}, aggregated along their respective axes, and finally fused via element-wise averaging to yield the 2D feature map. This strategy enables the 2D queries to adaptively integrate information from orthogonal axial tokens.

Reconstruction Objectives. We employ intensity-weighted L1 reconstruction as the primary training objective, complemented by two axis-specific auxiliary losses. While the reconstruction loss minimizes pixel-wise prediction errors, the auxiliary terms encourage the preservation of range- and azimuth-profile intensity statistics. The overall pre-training objective is formulated as

$$\mathcal{L}_{total} = \mathcal{L}_{rec} + \lambda_A \mathcal{L}_A + \lambda_R \mathcal{L}_R. \tag{3}$$

Here, $\mathcal{L}_{rec}$ denotes the base intensity-weighted mean absolute error (L1) loss for stripe reconstruction. The weighting emphasizes stronger measured responses without assuming that all such responses correspond to true targets. Meanwhile, $\mathcal{L}_A$ and $\mathcal{L}_R$ are auxiliary distribution-matching terms for the azimuth and range axes, respectively, with coefficients λ_A and λ_R.

Table 1: Tokenization ablation. For all configurations, MSE loss, random masking, and 500 epochs are applied identically. GFLOPs are measured during pre-training. Best results are shown in bold.

Patch size	# Tokens	$AP_{\mathrm{BEV}}(\%)\uparrow$	Encoder GFLOPs↓	Total GFLOPs↓
16×16	112	49.46	**17.67**	**23.97**
8×8	448	54.91	48.51	75.47
4×16	448	55.23	50.04	77.93
16×4	448	54.59	47.65	74.15
Axial	142	**57.46**	22.22	30.14

Azimuth-Profile Moment Loss. As depicted in Fig. 3, the finite aperture of the antenna array produces structured sidelobe responses along the azimuth axis [45, 48]. These responses can exhibit sharp peaks and asymmetric intensity distributions. To encourage consistency in their statistical characteristics, we match the variance (σ^2), skewness (g_1), and kurtosis (g_2) of the predicted and target intensities for masked azimuth tokens:

$$\begin{aligned}\mathcal{L}_A = &|\sigma^2(y) - \sigma^2(\hat{y})| \\ &+ |g_1(y) - g_1(\hat{y})| \\ &+ \lambda_k |g_2(y) - g_2(\hat{y})|.\end{aligned} \tag{4}$$

Here, y and $\hat{y}$ denote the target and reconstructed intensities, respectively. This moment-matching objective regularizes the distributional characteristics of azimuth responses but does not uniquely constrain the spatial locations, spacing, or periodicity of sidelobe peaks.

Range-Profile Distribution Loss. As formulated in Sec. 3.2, radar waves frequently undergo indirect reflections, such as those from vehicle undercarriages or surrounding structures [32–34]. These delayed reflections can produce asymmetric, heavy-tailed intensity profiles along the range axis. To encourage consistency in the empirical intensity distributions of reconstructed range profiles, we employ a 1D Wasserstein-based distribution loss [49]. For equally weighted scalar intensity samples, the 1D Wasserstein distance can be computed as the mean absolute difference between sorted samples:

$$\mathcal{L}_R = \frac{1}{N_R} \sum_{i=1}^{N_R} \left| \mathrm{sort}(y)_i - \mathrm{sort}(\hat{y})_i \right|. \tag{5}$$

Here, N_R denotes the number of scalar intensity samples from the masked range profiles, while y and $\hat{y}$ represent the target and reconstructed intensities, respectively. The $\mathrm{sort}(\cdot)$ operator is differentiable almost everywhere and has a computational complexity of $\mathcal{O}(N_R \log N_R)$. Because the loss compares sorted intensities rather than their range coordinates, it encourages distributional consistency without explicitly constraining the spatial locations or directions of multipath reflections.

4.4 Fine-Tuning for Object Detection

After pre-training, we discard the decoder and fine-tune the encoder with an anchor-based detection head identical to RTNH [2]. Axial tokens are projected to a BEV feature map through a lightweight stripe-to-BEV mapping, followed by ASPP-Lite [50] for multi-scale context.

5 Experiments

5.1 Datasets and Experimental Settings

K-Radar Dataset. The K-Radar dataset is a large-scale, widely used 4D radar dataset for 3D object detection in autonomous driving, providing 4D radar tensors (4DRT) that contain power measurements along the Doppler, range, azimuth, and elevation dimensions [2]. In this work, we generate RA maps from the provided 3D range-azimuth-elevation (RAE) maps by averaging over the elevation dimension, and use the resulting RA maps as the sole input modality for both RadarMAE pre-training and downstream object detection. We focus on the sedan class within a longitudinal range of $[0, 72]\,\mathrm{m}$ and a lateral range of $[-6.4, 6.4]\,\mathrm{m}$.

Evaluation Metrics. We evaluate detection performance using the intersection-over-union (IoU)-based average precision (AP) under the standard KITTI protocol [51] with 40 recall positions. We

Table 2: Comparison on K-Radar-20. BEV and 3D AP are reported at IoU 0.3/0.5. Methods marked with * use adapted backbones for the corresponding radar input. RadarMAE-Scratch shares the same architecture as RadarMAE but is trained without self-supervised pre-training. Best results are shown in bold.

Method	Backbone	SSL	Input	$AP_{\text{BEV}}(\%)\uparrow$		$AP_{\text{3D}}(\%)\uparrow$	
				AP@0.3	AP@0.5	AP@0.3	AP@0.5
EchoFusion [7]	*ResNet-50	✗	2D (RA)	54.45	42.28	51.75	24.65
RTNH [2]	3D-SCB	✗	3D (RAE)	61.38	46.47	53.05	17.98
ViT-Scratch [52]	ViT-B	✗	2D (RA)	59.49	44.16	55.24	27.64
MAE [17]	ViT-B	✓	2D (RA)	56.88	42.86	52.97	26.47
RadarMAE-Scratch	*ViT-B	✗	2D (RA)	60.77	45.30	55.24	27.45
RadarMAE	*ViT-B	✓	2D (RA)	**64.17**	**48.44**	**58.85**	**30.93**

report the AP for both BEV predictions (AP_{BEV}) and 3D bounding box predictions (AP_{3D}) on the sedan class, at IoU thresholds of 0.3 and 0.5 to assess the spatial localization accuracy of the model.

5.2 Effect of Axial Tokenization in MAE

To analyze the effect of the tokenization scheme in MAE, we first conducted a series of experiments with varying patch sizes, as shown in Table 1. Decreasing the isotropic square patch size from 16×16 to 8×8 yields a substantial improvement in AP_{BEV}. This indicates that finer resolution can better capture the physical signatures of sparse radar signals, albeit at the cost of an increased computational burden. To accommodate the distinct resolution disparities across the RA map axes and to encompass wider sidelobe regions within a single token, we applied asymmetric 4×16 patches, observing a further slight performance increase. Motivated by these findings, we introduced our proposed axial tokenization, which significantly boosts the performance to the highest AP_{BEV} of 57.46%. This confirms that axial tokenization is highly meaningful for effectively recognizing axis-specific resolutions and preserving the spatial continuity of extended sinc-shaped sidelobes without fragmentation. Furthermore, decoupling the spatial axes drastically reduces the total number of tokens from 448 in the 8×8 configuration to 142. As a result, RadarMAE requires less than half the pre-training computational cost compared to the asymmetric MAE variant, while delivering superior object detection accuracy.

5.3 Comparison with Baseline Methods

In this setup, we conduct experiments on the K-Radar-20 subset, consisting of the first 20 sequences of K-Radar, to align the dataset coverage with EchoFusion [7], which reports radar-only RA-map detection results on this subset. Additional performance comparisons on the full K-Radar dataset, comprising all 58 sequences, are presented in Table 6 in Appendix D.

Standard MAE vs. RadarMAE. We further investigate the impact of directly applying a standard vision-centric MAE to radar data. As shown in Table 2, the standard MAE baseline consistently underperforms across all spatial evaluation metrics compared to the ViT-B model trained from scratch without SSL. This notable drop indicates that directly applying image-oriented masked modeling to radar perception is ineffective and can even be detrimental, primarily due to the substantial domain gap and distinct physical characteristics of radar signals. In contrast, RadarMAE successfully overcomes this limitation by explicitly incorporating radar-specific physical priors. Despite being built upon a standard ViT architecture, our proposed framework achieves the best overall performance, reaching a peak of 64.17% in AP_{BEV}@0.3

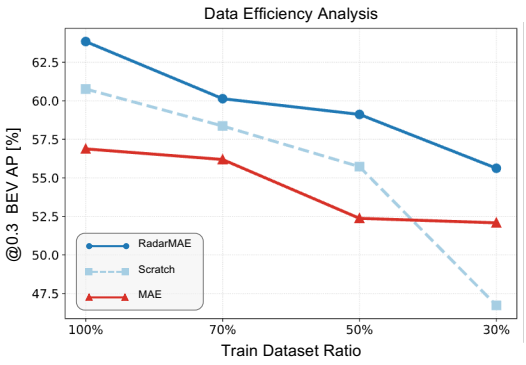


Figure 5: Data efficiency comparison using BEV AP@0.3.

and 30.93% in $AP_{\mathrm{3D}}@0.5$. These results support the effectiveness of incorporating radar-specific physical priors into self-supervised pre-training to learn representations that benefit downstream radar object detection.

Comparison with Other Radar Object Detection Methods. To further evaluate the effectiveness of RadarMAE for radar object detection, we compare it with representative baselines on the K-Radar-20 dataset, as shown in Table 2. To distinguish the effect of self-supervised pre-training from architectural differences, we include RadarMAE-Scratch, which shares the same axial tokenizer, PPE, encoder, stripe-to-BEV mapping, and detection head as RadarMAE but is trained from random initialization. Under this architecture-matched comparison, RadarMAE improves $AP_{\mathrm{BEV}}@0.3$ and $AP_{\mathrm{3D}}@0.3$ by 3.40 and 3.61 points, respectively. These results demonstrate the benefit of radar-aware pre-training beyond the architectural design alone.

Among existing detection methods, our primary baseline is the single-modality 2D RA configuration of EchoFusion [7], which provides a relevant input-matched comparison without camera, LiDAR, or temporal information. Compared to EchoFusion, RadarMAE achieves absolute improvements of 9.72 and 7.10 points in $AP_{\mathrm{BEV}}@0.3$ and $AP_{\mathrm{3D}}@0.3$, respectively. These results demonstrate the effectiveness of the proposed framework for radar object detection using dense RA heatmaps.

We further compare RadarMAE with RTNH [2], a widely used baseline that processes 3D radar RAE tensors. Although both methods employ the same anchor-based detection head, their input representations and encoders differ. Despite relying solely on 2D RA heatmaps without explicit elevation information, RadarMAE achieves higher performance across all reported metrics, including $AP_{\mathrm{3D}}@0.5$ of 30.93% compared with 17.98% for RTNH. This result suggests that radar-aware representation learning can extract informative spatial features for 3D object detection even from a lower-dimensional radar input, highlighting the potential importance of effective radar encoding beyond input dimensionality alone.

5.4 Ablation Studies

Comparison by Training Dataset Ratio. A primary motivation for adopting SSL is to improve data efficiency, which is particularly critical in the radar domain due to the immense difficulty and high cost of manual annotation. To thoroughly evaluate this, we conduct further experiments by varying the available training dataset ratio from 100% down to 30%, as illustrated in Fig. 5. Comparing the standard MAE with the model trained from scratch reveals a notable observation: when labeled data is extremely scarce, specifically at the 30% ratio, the standard MAE provides a distinct pre-training advantage over the scratch baseline. However, as the volume of training data increases, this benefit diminishes, and the standard MAE paradoxically underperforms the scratch model. This inversion implies that misaligned vision-centric priors in standard MAE ultimately hinder the network from fully leveraging larger amounts of radar data. Conversely, RadarMAE exhibits superior data efficiency across all scenarios. By correctly embedding radar-specific physical priors, RadarMAE not only prevents the degradation observed in standard MAE but also consistently maintains a significant performance gap over the scratch baseline across all data availability regimes.

Table 3: Ablation study on reconstruction loss formulations. All metrics are evaluated at IoU 0.3.

Loss	$AP_{\mathrm{BEV}}(\%)\uparrow$
$L2$	57.66
$L1$	62.85
$L1_w$	63.17
$\mathcal{L}_{total}$	64.17

Effect of Physical Distribution-Based Loss. To evaluate the effect of our proposed loss function, we compare different reconstruction losses in Table 3. The standard L2 loss (*i.e.,* MSE) yields the lowest performance (57.66% AP_{BEV}). Switching to an L1 objective substantially improves performance to 62.85%, suggesting that the choice of reconstruction objective is important for radar representation learning. Applying an intensity-weighted strategy ($L1_w$) further improves AP_{BEV} to 63.17%. Finally, incorporating the axis-specific physical distribution-based losses into the reconstruction objective ($\mathcal{L}_{\mathrm{total}}$) achieves 64.17% AP_{BEV}, providing an additional gain of 1.00 AP point over intensity-weighted L1. These auxiliary losses encourage consistency in the intensity statistics of range and azimuth profiles rather than directly constraining the spatial locations of multipath reflections or sidelobe peaks. Overall, these results indicate that the primary reconstruction objective accounts for a substantial portion of the performance improvement, while the physical distribution-based losses provide additional benefits as auxiliary regularizers.

Table 4: Free-space segmentation performance on RADIal. FFT-RadNet uses ADC-derived radar input and is included as a task-level reference. Best performance is shown in bold.

Method	Input	mIoU (%)↑
MAE [17]	RA map	73.03
RadarMAE-Scratch	RA map	74.85
FFT-RadNet [53]	ADC-derived	74.62
RadarMAE	RA map	**75.38**

5.5 Evaluation on RADIal Free-Space Segmentation

To evaluate the applicability of RadarMAE beyond K-Radar object detection, we conduct free-space segmentation experiments on RADIal [53], which differs in radar hardware, acquisition settings, and downstream task. While K-Radar uses the RETINA-4ST radar, RADIal employs a high-definition radar with 12 transmitters, 16 receivers, and 192 virtual antennas. For this evaluation, RadarMAE is independently pre-trained on RADIal RA maps and subsequently fine-tuned for free-space segmentation, without transferring pre-trained weights from K-Radar. This setting allows us to examine whether radar-aware self-supervised pre-training remains beneficial under a different sensor configuration and downstream task. As shown in Table 4, RadarMAE achieves the highest mIoU among the evaluated methods at 75.38%, outperforming vanilla MAE by 2.35 points and its architecture-matched scratch baseline by 0.53 points. The latter comparison helps distinguish the benefit of self-supervised pre-training from that of the architectural design. We also include FFT-RadNet as a task-level reference rather than an input-matched baseline, since it processes ADC-derived range-Doppler observations instead of RA maps. These results provide additional evidence of the applicability of radar-aware self-supervised pre-training to a different radar acquisition setting and downstream segmentation task.

6 Conclusion

In this paper, we proposed **RadarMAE**, a radar-aware MAE framework designed to incorporate the physical characteristics of FMCW radar signals. While conventional MAEs rely on vision-oriented isotropic patching and pixel-wise reconstruction, RA heatmaps exhibit anisotropic spatial structures and axis-specific physical artifacts that motivate radar-aware modeling. To bridge this domain gap, RadarMAE reformulates masked representation learning around the physical measurement structure of radar signals. At its core, axial tokenization represents RA heatmaps as orthogonal range- and azimuth-aligned sequences, accommodating resolution disparities while capturing spatially extended radar responses. Complementing this representation, intensity-weighted block masking emphasizes informative radar responses, while axis-specific auxiliary distribution-based losses encourage consistency in range- and azimuth-profile intensity statistics during reconstruction. Experiments on K-Radar-20 using only single-frame RA maps demonstrate that RadarMAE outperforms both conventional MAE and its architecture-matched scratch baseline, supporting the effectiveness of radar-aware self-supervised pre-training for downstream object detection. Overall, these findings highlight the value of incorporating radar-specific physical inductive biases into self-supervised learning to obtain effective representations for downstream radar perception.

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A Limitations and Future Works

Despite these promising results, RadarMAE currently focuses on representation learning from 2D RA heatmaps. This choice provides a computationally efficient input representation, but it does not explicitly exploit elevation information. A natural future direction is to extend RadarMAE to higher-dimensional radar representations, such as RAE heatmaps. Such an extension could better capture vertical structures and object geometry that are not fully represented in 2D RA heatmaps, potentially leading to improved detection performance at the cost of increased computational complexity.

Although we additionally evaluate free-space segmentation on RADIal, our evaluation remains limited to two radar datasets and a small number of downstream tasks. Further evaluation on broader sensor configurations and tasks, including object tracking, is needed to establish the generality of the learned representations.

In addition, incorporating Doppler information is another promising direction for improving the discrimination of moving objects and stationary clutter, particularly when they exhibit similar intensity patterns. The axis-specific auxiliary losses constrain aggregate intensity statistics rather than the exact locations of multipath returns or sidelobe peaks. Moreover, intersecting axial stripes and intensity-dependent masking can permit partial information recovery during pre-training. These limitations motivate further study of radar representation learning across broader settings.

B Broader Impacts

This work may improve radar-based perception for autonomous driving, particularly under adverse weather and low-visibility conditions where camera- or LiDAR-based systems can degrade. By using self-supervised pre-training, RadarMAE may also reduce reliance on costly manual annotations for radar heatmaps. At the same time, radar perception is safety-critical. Improved benchmark performance does not guarantee safe deployment, and failures such as missed detections, false positives, or poor generalization could affect downstream driving decisions. Thus, RadarMAE should be viewed as a representation learning framework that requires extensive real-world validation before deployment. More capable radar sensing may also raise privacy and misuse concerns, since radar can reveal the presence, motion, and spatial behavior of people and vehicles. Responsible use should include appropriate data governance and compliance with relevant safety and privacy regulations. Finally, although the proposed 2D RA-based design is relatively efficient, future extensions to higher-dimensional radar inputs may increase computational cost, making energy efficiency and deployment practicality important considerations.

C Implementation Details

Pre-training. We pre-train a ViT-Base encoder with an embedding dimension of 768, 12 Transformer layers, 12 attention heads, and an MLP ratio of 4.0. The input RA maps are generated from K-Radar and resized to 112×258 (azimuth$\times$range) with a single channel. The radar intensity values are converted to the dB scale and normalized using dataset-level statistics. We adopt the proposed axial stripe tokenizer and use physical positional encoding (PPE) based on Fourier features. During pre-training, we apply intensity-weighted block masking with a nominal masking ratio of 0.75, contiguous blocks of 5–8 tokens, and a randomly retained visible-hint token per selected block. The model is optimized using a patch-intensity-weighted ℓ_1 reconstruction loss together with auxiliary azimuth-profile moment and range-profile distribution losses, with both auxiliary loss weights set to 0.1. Pre-training is performed for 800 epochs using AdamW [54] with a base learning rate of 1×10^{-5}, weight decay of 0.05, a 50-epoch warm-up, and a cosine learning-rate schedule.

Fine-tuning. For downstream 3D object detection, we initialize the encoder with the pre-trained RadarMAE weights and attach it to an RTNH-style anchor-based detection pipeline with polar-BEV feature refinement. The region of interest is set to $[0, 72] \times [-6.4, 6.4] \times [-2.0, 7.6]\,\mathrm{m}$, with a voxel size of $0.4\,\mathrm{m}$ and an azimuth field of view of $\pm 53^\circ$. We evaluate the *Sedan* class using K-Radar v1.0 annotations under the KITTI evaluation protocol. The detector is fine-tuned using AdamW with an initial learning rate of 1×10^{-4}, a minimum learning rate of 1×10^{-5}, weight decay of 10^{-3}, batch size 8, and cosine annealing for 11 epochs.

Compute resources. The primary K-Radar benchmark experiments were conducted on a workstation equipped with NVIDIA H200 GPUs (140 GB HBM3e per GPU). The factorial ablation in Appendix I was conducted on a separate GPU server using the same codebase and evaluation protocol. Pre-training of the ViT-Base RadarMAE model for 800 epochs on the full K-Radar pre-training set is performed on a single H200 GPU and takes approximately 9 hours, with a peak GPU memory consumption of about 17 GB. Fine-tuning the RTNH-style detector for 11 epochs on a single H200 GPU takes approximately 0.5 hours per run and uses about 7 GB of GPU memory. The full set of experiments reported in the main paper and the supplementary material, including pre-training ablations across tokenizer, positional encoding, masking, and loss configurations, downstream fine-tuning runs across seeds and ablations, and downstream evaluation, amounts to roughly 200 GPU-hours on a single H200. Beyond the runs reported in the paper, the overall research project required additional compute for preliminary experiments, hyperparameter exploration (e.g., masking ratio, patch size, loss weights, learning-rate schedules), and design iterations on the tokenizer and positional encoding, which we estimate to be approximately 2–3$\times$ the compute of the final reported experiments.

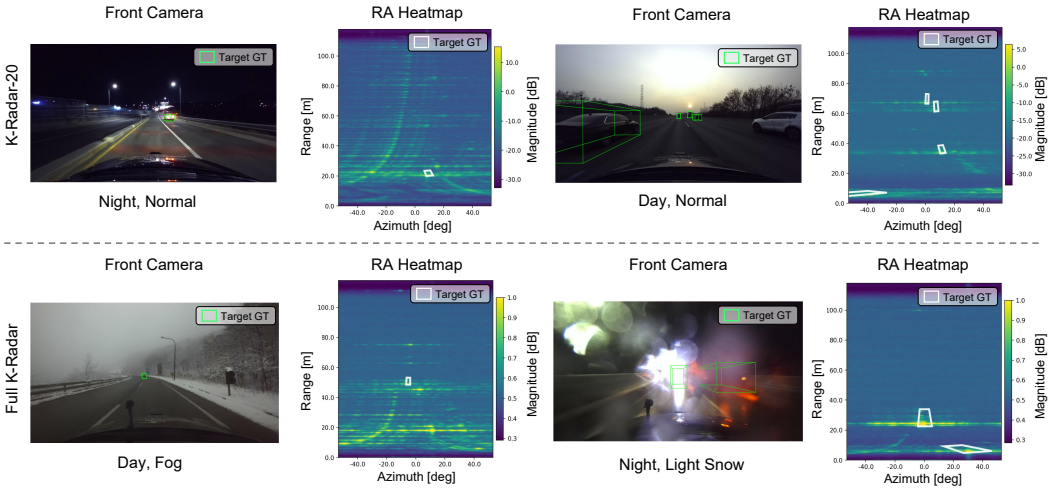


Figure 6: Qualitative examples of front-camera images and corresponding RA heatmaps from K-Radar-20 and the full K-Radar dataset. K-Radar-20 mainly contains normal-weather scenes, whereas the full K-Radar dataset includes more challenging adverse weather conditions such as fog and light snow. Green boxes indicate target ground-truth annotations in the camera view and their corresponding locations in the RA heatmaps.

D Performance on the Full K-Radar Dataset

Table 5: Comparison between K-Radar-20 and the full K-Radar dataset.

Group	Category	K-Radar-20		Full K-Radar	
		Seq.	Frames	Seq.	Frames
Scale	Total	20	13,008	58	34,994
Time	Day	13	9,008	35	21,920
	Night	7	4,000	23	13,074
Weather	Normal	19	12,781	19	12,781
	Overcast	1	227	2	825
	Rain	0	0	8	4,780
	Sleet	0	0	12	7,040
	Fog	0	0	6	3,578
	Light snow	0	0	4	2,237
	Heavy snow	0	0	7	3,753

To clarify the difference between K-Radar-20 and the full K-Radar dataset, we summarize their scale, time-of-day distribution, and weather diversity in Table 5. K-Radar-20 contains 20 sequences and 13,008 frames, most of which are collected under normal weather conditions. In contrast, the full K-Radar dataset contains 58 sequences and 34,994 frames, covering diverse adverse weather conditions such as rain, sleet, fog, light snow, and heavy snow [2]. Fig. 6 provides qualitative examples showing that the full dataset includes more challenging weather scenes where camera observations are degraded while RA heatmaps remain informative.

We further evaluate RadarMAE on the full K-Radar dataset to verify its scalability under broader data distributions. As shown in Table 6, RadarMAE achieves the best performance across the reported BEV and 3D detection metrics. It outperforms the scratch ViT-B and standard MAE baselines, providing further evidence of the effectiveness of the radar-aware design beyond the K-Radar-20 subset. RadarMAE also surpasses RTNH despite using only 2D RA maps instead of 3D RAE inputs, although these methods use different input representations and architectures.

Table 6: Performance comparison on the full K-Radar dataset. Best results are shown in bold.

Method	Backbone	SSL	Input	$AP_{\mathrm{BEV}}(\%)\uparrow$		$AP_{\mathrm{3D}}(\%)\uparrow$	
				AP@0.3	AP@0.5	AP@0.3	AP@0.5
RTNH [2]	3D-SCB	✗	3D (RAE)	58.04	42.60	49.65	17.87
ViT-B [52]	ViT-B	✗	2D (RA)	58.45	44.18	53.29	28.99
MAE [17]	ViT-B	✓	2D (RA)	55.32	42.85	51.99	28.00
RadarMAE	*ViT-B	✓	2D (RA)	**60.48**	**47.76**	**56.43**	**29.04**

E Qualitative Results

E.1 Object Detection Results

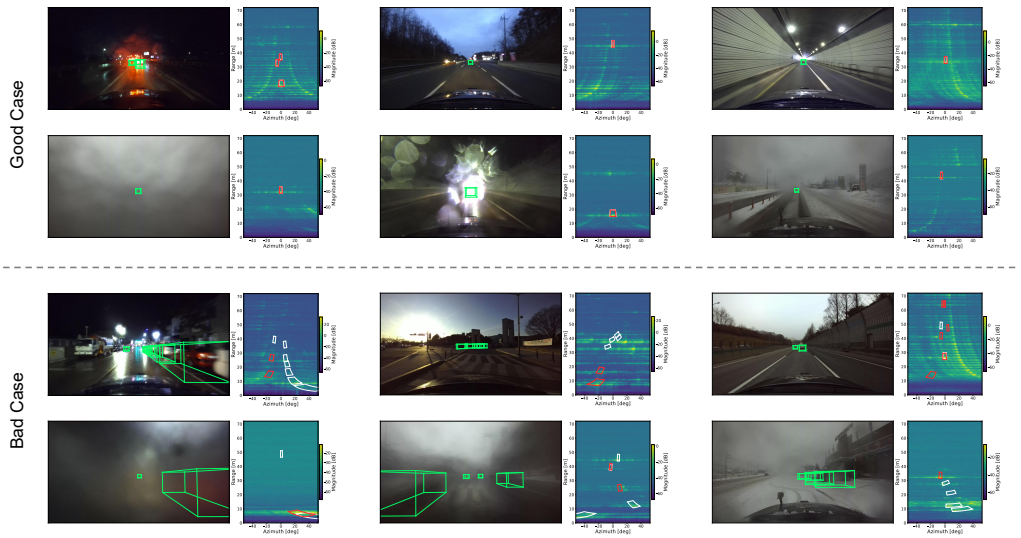


Figure 7: Qualitative detection results of RadarMAE on challenging driving scenes. The top group shows representative good cases, where RadarMAE accurately localizes target vehicles from single-frame RA heatmaps despite adverse visual conditions such as nighttime, fog, tunnels, and snowy roads. The bottom group shows representative bad cases, where weak target responses, severe clutter, or ambiguous radar reflections lead to missed detections or inaccurate localization. In the camera view, green boxes denote ground-truth annotations. In the RA heatmaps, white boxes denote ground-truth target regions, while red boxes denote the predictions of RadarMAE.

E.2 Visualization of Occlusion-Based Sensitivity Maps

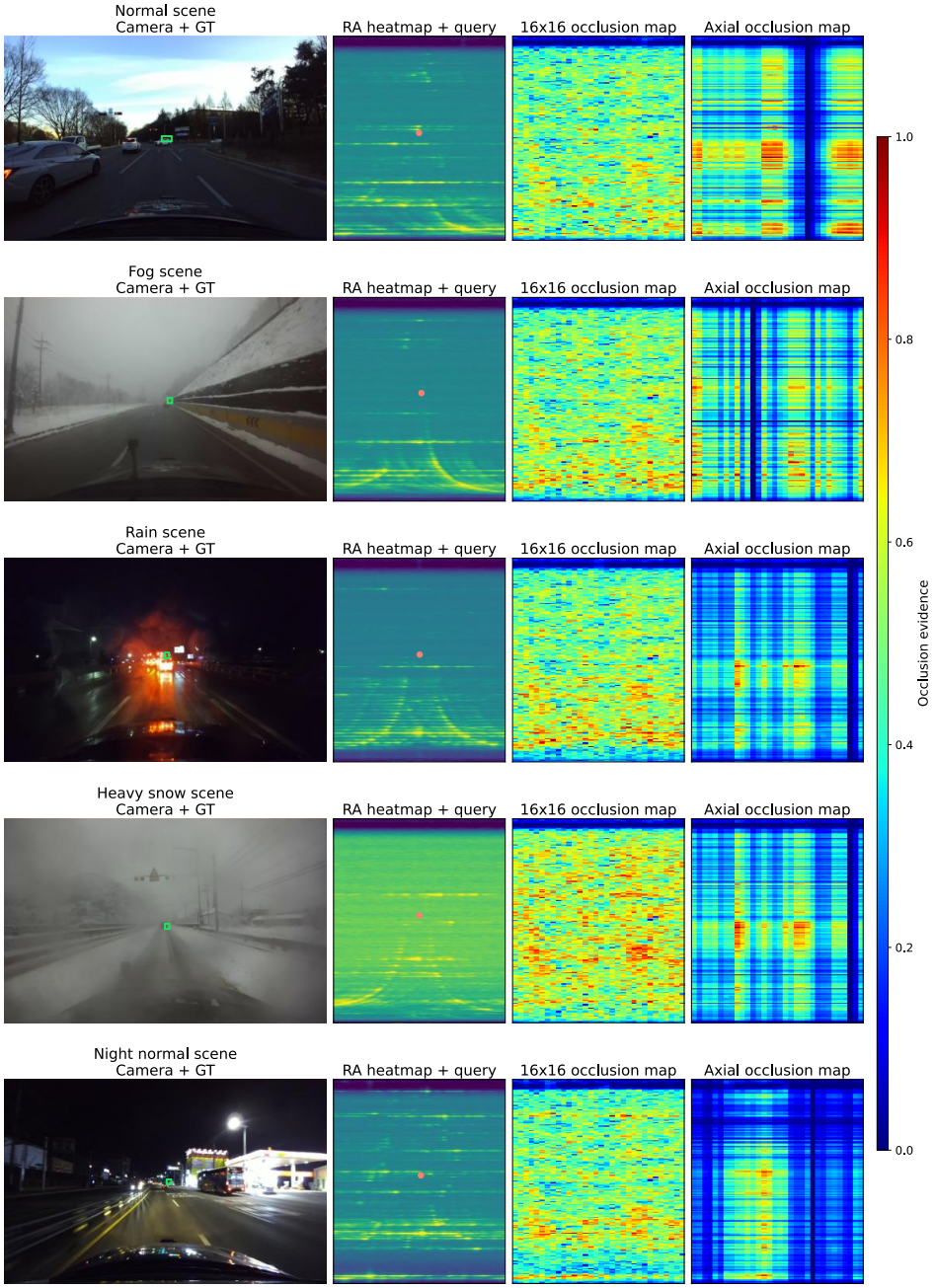


Figure 8: Visualization of occlusion-based attention evidence under diverse driving conditions. Each row shows a front-camera image with the ground-truth target, the corresponding RA heatmap with the query location, the occlusion map obtained from conventional 16×16 square patches, and the occlusion map obtained from the proposed axial tokenization. Compared with the noisy and spatially fragmented responses of square patch occlusion, axial occlusion produces more structured axis-aligned evidence around target-related radar signatures under the evaluated conditions.

F Weighted Block Masking

The intensity-weighted block masking procedure and visible hint mechanism are illustrated in Fig. 4 in the main paper. As shown in Table 7, weighted block masking outperforms random masking across all reported BEV and 3D detection metrics. In particular, it improves $AP_{\mathrm{BEV}}@0.3$ from 60.95% to 64.17% and $AP_{\mathrm{3D}}@0.5$ from 28.22% to 30.93%. These results support the effectiveness of the proposed masking strategy under the evaluated protocol.

Table 7: Comparison of masking strategies on K-Radar-20. BEV and 3D AP are reported at IoU 0.3/0.5. Best results are shown in bold.

Masking	$AP_{\mathrm{BEV}}(\%)\uparrow$		$AP_{\mathrm{3D}}(\%)\uparrow$	
	$AP@0.3$	$AP@0.5$	$AP@0.3$	$AP@0.5$
Random	60.95	44.98	55.54	28.22
Weighted block	**64.17**	**48.44**	**58.85**	**30.93**

Controlled Mask Selection Comparison. To further isolate the effect of intensity-weighted mask selection, we conduct a separate comparison between uniform random and intensity-weighted selection while keeping the block size, masking ratio, hint-token policy, architecture, and training protocol unchanged. Intensity-weighted selection improves $AP_{\mathrm{BEV}}@0.3$ from 61.79% to 62.53%, corresponding to a gain of 0.74 AP points. This result suggests that intensity-aware selection provides an additional benefit under matched conditions, although intensity-dependent masking may still reveal partial information about the input signal.

G Positional Embedding

We further analyze the effect of positional encoding by comparing conventional positional embedding (PE) with the physical positional encoding (PPE) described in Sec. 4. In this experiment, all components of the final RadarMAE framework are kept fixed, and only the positional encoding strategy is changed. As shown in Table 8, PPE improves both BEV and 3D detection performance under the evaluated protocol, suggesting that measurement-coordinate-based embeddings provide useful spatial cues for downstream localization.

Table 8: Comparison of positional embedding strategies on K-Radar-20. BEV and 3D AP are reported at IoU 0.3/0.5. Best results are shown in bold.

Positional Embedding	$AP_{\mathrm{BEV}}(\%)\uparrow$		$AP_{\mathrm{3D}}(\%)\uparrow$	
	$AP@0.3$	$AP@0.5$	$AP@0.3$	$AP@0.5$
PE	61.23	45.04	56.19	28.81
PPE	**64.17**	**48.44**	**58.85**	**30.93**

H Fine-Tuning Details

Feature Reshaping. After the self-supervised pre-training, we discard the decoder and initialize the encoder with the learned weights to adapt the representations for the downstream detection task. To construct a dense spatial feature map from the axial tokens, we employ a direct stripe-to-BEV mapping that projects the axial token bank onto a 2D BEV grid. During this projection, geometric alignment is achieved by generating polar reference points based on the BEV query coordinates, where learned offsets explicitly compensate for the nonlinear coordinate transformations between the radar polar space and the Cartesian BEV space. Finally, a lightweight ASPP-Lite block [50] is applied to the projected BEV feature map to enrich multi-scale contextual information without heavy computational overhead.

Detection Head. We adopt a standard anchor-based detection head identical to the RTNH baseline [2]. The entire network is fine-tuned end-to-end using a joint detection loss $\mathcal{L}_{\text{det}}$, which is a combination

of focal loss for classification, smooth-L1 loss for bounding box regression, and a directional classification loss.

Table 9: Factorial ablation of RadarMAE components on K-Radar-20. All configurations use the same primary L1 reconstruction objective and evaluation protocol. Checkmarks indicate enabled components, while blank cells indicate disabled components. Best performance is shown in bold.

Axial Tokenization	Block Masking	Physical Loss	AP_{BEV}@0.3 (%)↑
			61.79
		✓	61.93
	✓		62.53
	✓	✓	63.74
✓			64.78
✓		✓	64.53
✓	✓		65.16
✓	✓	✓	**65.61**

I Factorial Ablation of RadarMAE Components

Table 9 presents a controlled $2 \times 2 \times 2$ factorial ablation of axial tokenization, block masking, and physical distribution-based losses. All eight configurations were evaluated on the same GPU server using the same codebase and evaluation protocol. Axial tokenization provides the largest individual improvement, increasing AP_{BEV}@0.3 from 61.79% to 64.78%. With axial tokenization enabled, block masking further improves performance to 65.16%, while adding the physical losses yields 65.61%. However, without block masking, adding the physical losses slightly decreases performance from 64.78% to 64.53%.

The factorial experiments were conducted on a different GPU server from the main benchmark. Accordingly, we interpret the results primarily through comparisons within the factorial experiment rather than directly comparing absolute AP values across the two experimental environments. These results identify axial tokenization as the primary contributor and suggest that the auxiliary losses have a configuration-dependent effect.

J Robustness to Radar Sampling Resolution

To evaluate the sensitivity of RadarMAE to radar sampling resolution, we independently reduce the range or azimuth sampling resolution by a factor of two, following the experiments reported in our rebuttal. As shown in Table 10, RadarMAE exhibits smaller performance degradation than both MAE and RadarMAE-Scratch. Specifically, its BEV AP@0.3 decreases by only 0.40 and 1.23 points under reduced range and azimuth sampling, respectively. These experiments assess robustness to sampling degradation rather than changes in physical antenna aperture or beamwidth.

Table 10: BEV AP@0.3 (%) under reduced range and azimuth sampling on K-Radar-20. Best performance is shown in bold.

Method	Original	Range/2	Azimuth/2
MAE	56.88	48.87	44.36
RadarMAE-Scratch	60.77	58.03	49.25
RadarMAE	**64.17**	**63.77**	**62.94**

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